The Moon – birth, craters, tides and life

4.5 billion years ago the young Earth was struck by a planet the size of Mars. The Moon formed from the wreckage – but why not a ring, like Saturn's? Follow the impact in 3D, build a moon or a ring yourself, watch a crater form in slow motion, and see how the Moon keeps Earth's axis steady and raises the tides that may have coaxed the first four-legged animals onto land.

Michael H. Grant · lektor emeritus in physics & mathematics · **mhgfysik.dk/maanens-dannelse-en.html** – the interactive 3D edition · CC BY 4.0 · November 2026

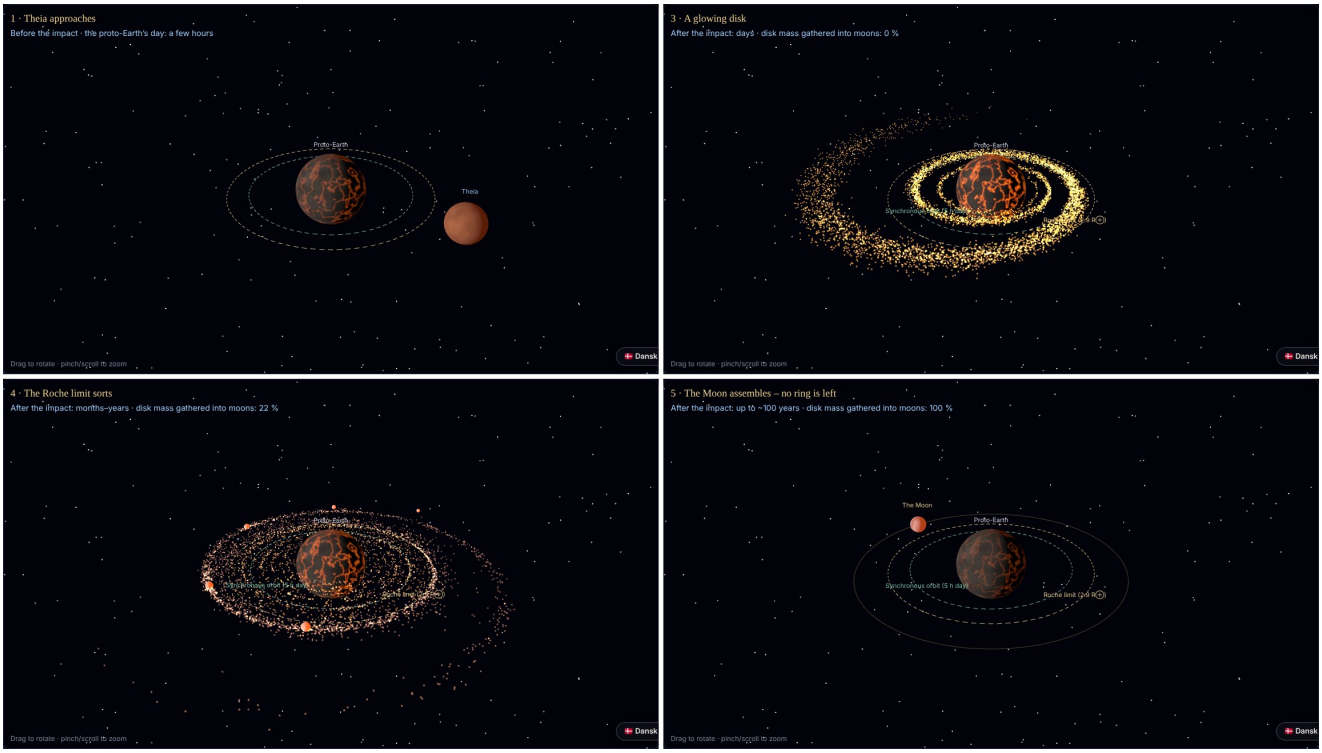


Figure 1. Four snapshots from the 3D animation "Formation": Theia approaches the proto-Earth; the glowing disk with the Roche limit (yellow) and the synchronous orbit (green); moonlets grow outside the Roche limit while the inner ring spreads; the moonlets merge into the Moon.

About the model: the time axis is compressed and non-linear. In the formation scene Earth and Moon are drawn to the correct relative size, and orbital speeds follow Kepler's third law ($T \propto r^{3/2}$), but slowed down. The number of particles and the thickness of the disk are stylised. The crater scene uses the scaling laws of Collins, Melosh & Marcus (2005).

1 · A giant impact made the Moon

Before the Apollo programme there were three old explanations: the Moon split off from a rapidly spinning Earth (George Darwin, 1879), was captured, or formed alongside Earth. None survived once the astronauts brought home 382 kg of moon rock. In 1975–76 William Hartmann and Donald Davis, and independently Alastair Cameron and William Ward, proposed that Earth had been struck by another protoplanet. It was later named **Theia**, after the Titan who in Greek mythology is the mother of the moon goddess Selene.

The theory explains four things at once:

- **The small iron core.** The Moon's core is only about 1–2 % of its mass (Earth's: 32 %). In the impact the iron sank into Earth; what was flung out was mostly mantle rock.
- **The dry, "boiled" composition.** Moon rocks lack volatile substances such as water, potassium and zinc – they boiled away in the hot disk.

- **The angular momentum.** The Earth–Moon system has an unusually large angular momentum – just what an oblique blow provides.
- **The magma ocean.** The Moon's bright highlands are made of anorthosite, which floated to the top of an ocean of molten rock – so the Moon was molten when it formed.

One thing was puzzling: Earth and Moon have almost identical isotopic compositions (oxygen, titanium, chromium, iron), even though the simulations say the Moon should be made mostly of Theia. That has led to variants: a more violent impact that mixed everything into a spinning cloud of vapour – a **synestia** (Lock & Stewart, 2018) – or that the Moon formed directly within hours (Kegerreis et al., 2022). A newer answer is that Theia was simply Earth's neighbour: iron isotopes indicate that Theia formed in the inner Solar System, perhaps even closer to the Sun than Earth (Hopp et al., *Science*, November 2025).

Remnants of Theia may still lie deep inside Earth: two huge regions in the lowermost mantle beneath Africa and the Pacific, where seismic waves travel slowly, have been proposed as Theia's buried mantle (Yuan et al., *Nature*, 2023).

The Moon in numbers	
Radius	1 737 km (0.273 R⊕)
Mass	$7.35 \cdot 10^{22}$ kg (1/81 of Earth's)
Mean density	3.34 g/cm³
Mean distance	384 400 km (60.3 R⊕)
Receding	3.8 cm/year
Sidereal / synodic month	27.32 / 29.53 days
Age	about 4.5 billion years

2 · Why didn't the Moon end up as a ring around Earth?

Right after the impact the debris really was a ring – a flat disk of molten rock and vapour around Earth. Saturn's rings show that such a ring can survive. So why did Earth's become a moon? The answer lies in two distances.

The Roche limit. Earth pulls harder on the side of a clump facing Earth than on the side facing away. The difference – the tidal force – tears the clump apart if it is stronger than the clump's own gravity. For a fluid or loosely bound body Édouard Roche found in 1849:

$$d_R = 2.44 \cdot R\oplus \cdot (\rho\oplus / \rho_m)^{1/3} = 2.44 \cdot (5.51 / 3.34)^{1/3}\, R\oplus \approx 2.9\, R\oplus \approx 18\,400\ km$$

Inside this distance, rock fragments can only form a ring; outside it they can clump together.

The synchronous orbit. Where one orbit lasts exactly one day. With Kepler's third law and Earth's day of about 5 hours just after the impact:

$$a_s = (GM\oplus\, T^2 / 4\pi^2)^{1/3} \approx 14\,900\ km \approx 2.3\, R\oplus$$

Simulations (Ida, Canup & Stewart, 1997; Salmon & Canup, 2012) then show what happened:

1. **Most of the mass lay outside the Roche limit.** An oblique blow sends material out to 3–6 R⊕. Moonlets grew there within months to years.
2. **The inner ring was unstable.** It was hot, dense and full of collisions. It spread out: its innermost part rained down on Earth, while the outermost part was pushed across the Roche limit and fed the moonlets. A ring of molten rock and vapour lives far shorter than Saturn's ice-cold rings – perhaps only ~100 years.

3 **The Moon was born outside the synchronous orbit.** Earth then spun faster than the Moon orbited, and the tidal bulge ran ahead of the Moon and pulled it *outwards* – away from the Roche limit. Had the Moon been born inside the synchronous orbit, it would instead have spiralled inwards and been torn into a ring. That is exactly the fate of Mars' moon Phobos: it orbits inside Mars' synchronous orbit and is expected to become a ring in 30–50 million years.

4 **Rock is dense.** Ice has a low density and a Roche limit far out (about 4.4 R⊕ at Earth). Saturn's rings are ice and lie inside Saturn's Roche limit. Try "An icy moon" in the laboratory.

See also Saturn's rings – birth, life and death and The Moon's orbit in 3D.

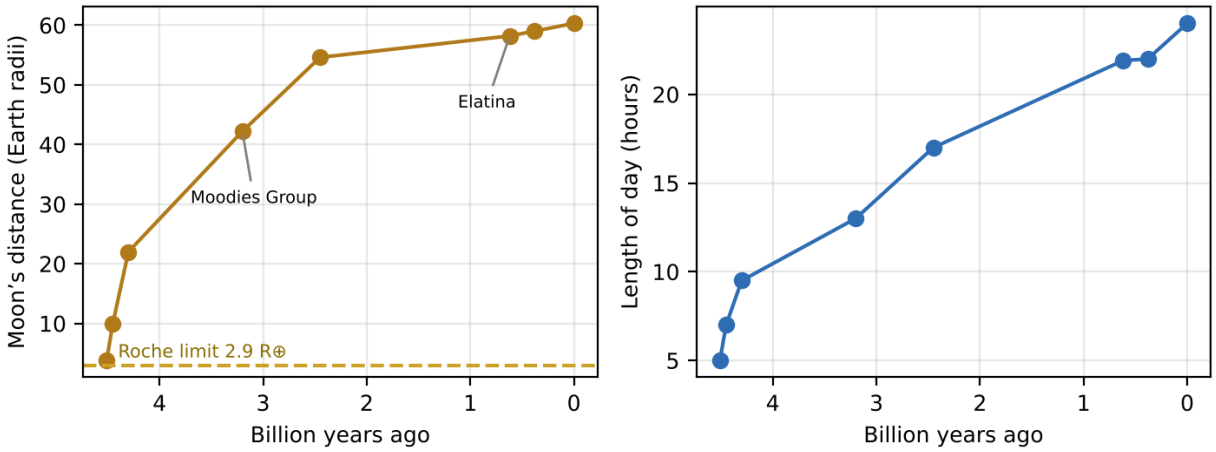


Figure 2. The Moon's distance and the length of day through Earth's history (approximate; the points at 3.2 billion and 620 million years rest on tidal deposits). The Moon formed just outside the Roche limit and has since receded to 60 Earth radii.

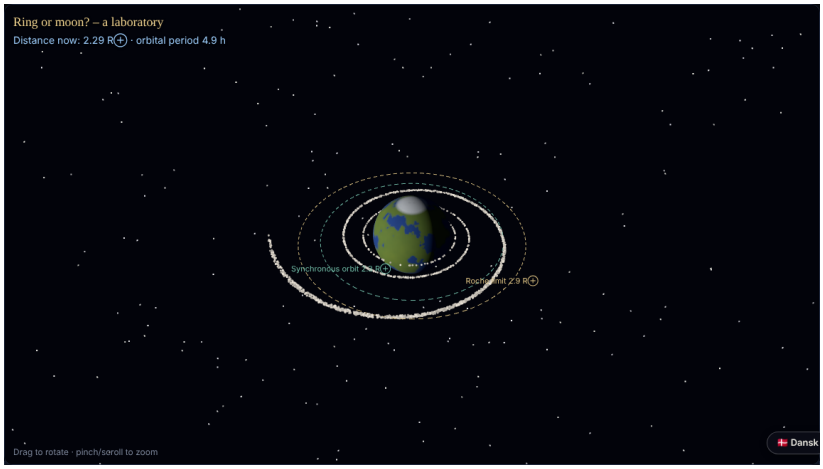


Figure 3. The laboratory "Ring or moon?": a Moon-sized rubble pile placed inside the Roche limit is torn apart and drawn out into a ring.

3 · How a crater forms

The Moon's near side is covered by around a million craters larger than 1 km. For a long time many thought they were volcanoes; Grove Karl Gilbert argued for impacts in 1893, but only in the 1960s – with Eugene Shoemaker's studies of Meteor Crater in Arizona and the craters of nuclear tests – did this become widely accepted. An impact at 15–20 km/s behaves like an explosion. It happens in three stages, which you can follow in the 3D scene "Crater":

1 **Contact and compression** – fractions of a second. A shock wave with pressures of millions of atmospheres runs into the ground; the impactor vaporises.

2 **Excavation** – seconds to minutes. Behind the shock wave the rock flows outwards and upwards. A bowl-shaped transient crater opens and a cone-shaped curtain of ejecta flies out.

3 **Modification** – minutes. Gravity takes over. Small craters become bowls with a lens of crushed rock; large ones get terraces and a central peak.

The crater size can be predicted with a scaling law. For the transient crater (Collins, Melosh & Marcus, 2005):

$$D_{tc} = 1.161\ (\rho_i/\rho_t)^{1/3}\ L^{0.78}\ v^{0.44}\ g^{-0.22}\ \sin^{1/3}\theta$$

Here L is the impactor diameter, v the speed, g the gravitational acceleration and θ the angle above horizontal (SI units). Note that g appears: the same impact makes a larger crater on the Moon than on Earth. A 1 km rocky body at 17 km/s gives a crater about 18 km across on the Moon – roughly 18 times its own diameter. Excavation lasts about $0.8\cdot\sqrt{(D_{tc}/g)}$, i.e. a little over a minute.

Craters are almost always round, even for oblique impacts; only below about 15° do they become elongated with butterfly-shaped ejecta. Well-known examples: Tycho (85 km) with its bright rays, Copernicus (93 km) with central peaks and terraces, Meteor Crater (1.2 km, about 50,000 years old) and Chicxulub (about 180 km), which ended the dinosaurs 66 million years ago.

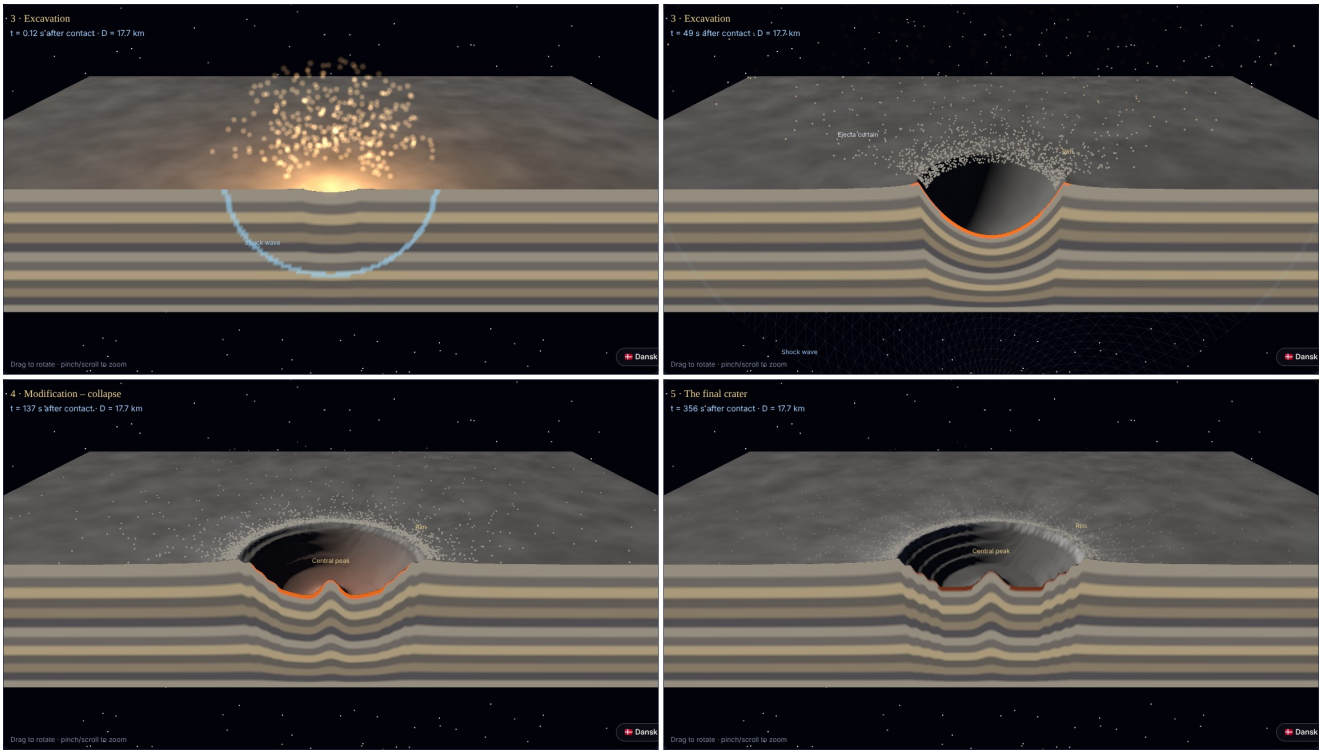


Figure 4. Crater formation in cross-section (1 km rocky body, 17 km/s, 45°, on the Moon): the shock wave (blue) in the contact and compression stage; excavation with the ejecta curtain; collapse, where the floor rebounds into a central peak; the final complex crater with terraces, about 18 km across.

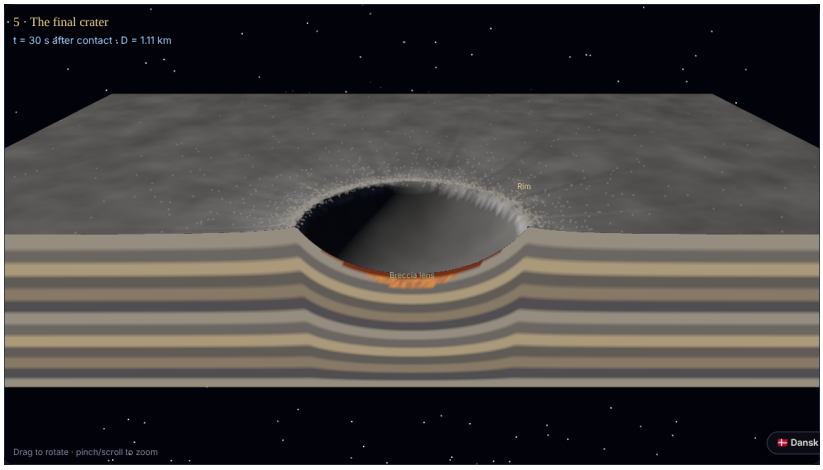


Figure 5. A simple bowl crater (the Meteor Crater scenario, about 1.1 km): the walls have slumped and filled the floor with a lens of crushed rock (breccia).

4 · Does the Moon protect Earth?

Yes – against a wobbling climate. Earth's axis is tilted 23.4°, which gives the seasons. The pull of the Sun and the Moon on the equatorial bulge makes the axis precess with a period of about 26,000 years. The other planets slowly perturb the plane of Earth's orbit. If the precession fell into step with those perturbations, the tilt could swing wildly. The Moon makes the precession fast enough to avoid the resonance, and the tilt only varies between about 22.1° and 24.5° over 41,000 years – enough to pace the ice ages, but not to turn the climate upside down.

Jacques Laskar and colleagues calculated in 1993 that without the Moon the tilt would wander chaotically between 0° and 85°. At 85° the poles would alternate half a year of scorching midnight sun with half a year of darkness. Mars, which has no large moon, has precisely such a chaotic tilt, which has swung between about 0° and 60°. Lissauer, Barnes & Chambers (2012), however, found that a moonless Earth spinning the same way as today would see more modest swings – typically 10–20° over a billion years. The conclusion: the Moon makes Earth's climate more stable, but how decisive that has been for life is still debated.

No – not as a shield against asteroids. This is a widespread myth. The Moon covers only a tiny fraction of the sky as seen from Earth. The chance that a random asteroid heading for Earth hits the Moon instead is roughly the ratio between the Moon's cross-section and a sphere with a radius equal to the Moon's distance:

$$\pi R_M^2 / (4\pi d^2) = (1737 / 384\,400)^2 / 4 \approx 5 \cdot 10^{-6}$$

The Moon's craters are not scars from blows it took on our behalf; they are an *archive* of the blows that also hit Earth, but which weather, water and plate tectonics have erased here.

And the Moon has slowed Earth down. Tidal friction has lengthened the day from about 5 to 24 hours. A planet with a 5-hour day would have stronger winds and different weather systems.

5 · The tides

The tidal force is the difference between the Moon's attraction at Earth's surface and at Earth's centre. With Earth's radius R and the Moon's distance d it is about

$$a_{tid} \approx 2GMR / d^3 = 1.1 \cdot 10^{-6}\ \text{m/s}^2\ \text{(Moon)},\ 0.5 \cdot 10^{-6}\ \text{m/s}^2\ \text{(Sun)}$$

Note the d^3: halving the distance gives tides eight times larger. The Sun is 27 million times more massive than the Moon, but 390 times farther away, so its tide is only 46 % of the Moon's. That the two are of the same order is connected with the Sun and Moon appearing the same size in the sky – as in a total solar eclipse.

At new and full moon the Sun and Moon pull in line: **spring tides**. At half moon they are at right angles: **neap tides**. So the tide beats with a period of half a synodic month, 14.8 days – see the curve in the 3D scene "Tides".

In the past the Moon was closer. Tidal deposits from the Moodies Group in South Africa show that 3.2 billion years ago the Moon was about 70 % of its present distance away, and that the day was about 13 hours (Eulenfeld & Heubeck, 2023). The tidal force was then $(1/0.7)^3 \approx 3$ times larger. In the Devonian, about 380 million years ago, the Moon was only 2–3 % closer than today, and the year had about 400 days, as John Wells (1963) counted in the daily growth lines of fossil corals.

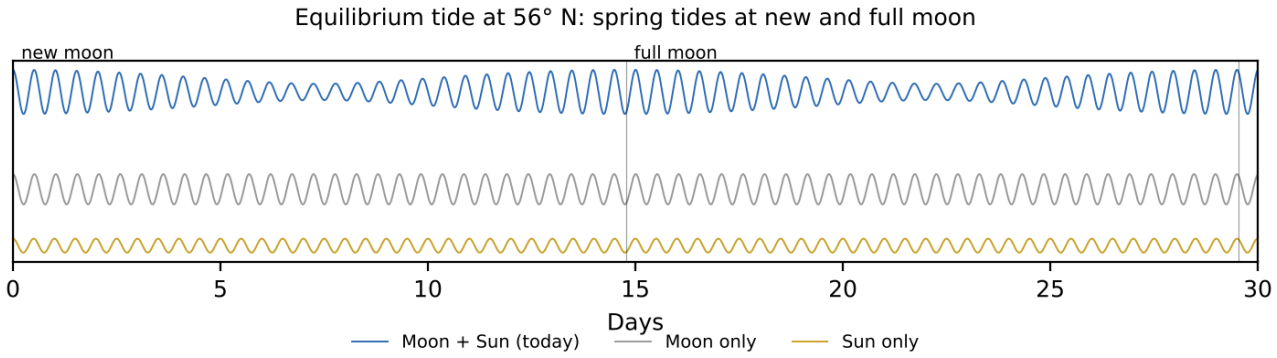


Figure 6. The lunar and solar tides have almost the same period (12.42 and 12.00 hours), so they beat: spring tides at new and full moon, neap tides at half moon.

6 · The tides and life's path onto land

Life began in the sea. The first plants and arthropods started moving onto land in the Ordovician and Silurian, about 470–420 million years ago, and in the Late Devonian, about 375 million years ago, fish with sturdy fins – like *Tiktaalik* – crawled into shallow water and out onto land. Here the intertidal zone plays a special role: a belt that switches between sea and land twice a day, where one can adapt to air a little at a time.

The astrophysicist Steven Balbus (Oxford, 2014) suggested that the Moon's tides were the driving force. Because the solar and lunar tides are almost equal but have slightly different periods, they beat strongly. At spring tides the sea reaches far up; at neap tides it can be days before it returns. In the long, narrow seaways of the Devonian this created a network of tidal pools that were cut off from the sea and dried up. A fish that could crawl from a drying pool to the next one, or back to the sea, survived. Detailed computer models of the Devonian oceans (Byrne et al., 2020) found tides of over 4 metres exactly where the earliest bony fish and four-legged animals have been found.

This is a hypothesis, not settled fact. Others point to drought, oxygen-poor water, or hunting for food in shallow swamps. But a world without the Moon would have small, regular solar tides and far fewer such pools.

7 · Moonlight – a clock for life in the sea

The full Moon gives only about 0.25 lux – some 400,000 times less than full sunlight. Even so, it is enough to set a biological clock. Many marine animals have a **circalunar** clock with a period of about 29.5 days, governed by light-sensitive proteins (cryptochromes) that can sense even faint moonlight.

- **Corals** spawn in synchronised mass releases a few nights after full moon. Experiments show that the cue is the dark period between sunset and moonrise that appears after full moon, because the Moon rises about 50 minutes later each evening (Lin et al., PNAS 2021).
- **Horseshoe crabs** – a lineage over 440 million years old – crawl up the beaches at high tide near new and full moon to lay their eggs as high up the beach as possible. A living picture of the border between sea and land.
- **Grunion** in California leap onto the beach and spawn in the sand at night just after spring tides – the eggs hatch at the next spring tide two weeks later.
- **Sandhoppers** and **dung beetles** navigate by the Moon and the polarisation of moonlight.

Moonlight and tides follow the same calendar, and for animals in the intertidal zone they are closely linked. There is no direct evidence, though, that moonlight itself drove life onto land; the tides are the stronger candidate.

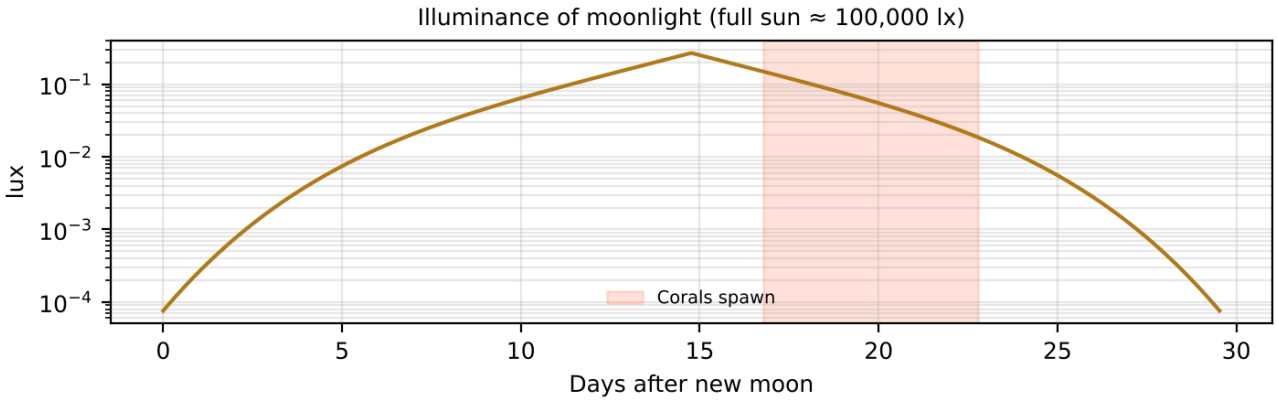


Figure 7. The illuminance of moonlight through a synodic month. The full Moon is more than ten times brighter than the half Moon. Corals typically spawn 2–8 nights after full moon, when the evenings begin with a dark period before moonrise.

Sources and further reading

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